

1) Calculate mean, median, mode for the following data.

Cental size	10	15	20	25	30	35	40	45
Frequency	10	25	40	50	25	17	8	5

sol:-

i) mean:-

Cental size x	Frequency f	fx
10	10	100
15	25	375
20	40	800
25	50	1250
30	25	750
35	17	595
40	8	320
45	5	225
$\Sigma f = 180$		$\Sigma fx = 4415$

$$\bar{x} = \frac{\Sigma fx}{N}$$

$$\bar{x} = \frac{4415}{180}$$

$$\bar{x} = 24.52 //$$

ii) median:-

x	f	cf
10	10	10
15	25	35
20	40	75
25	50	125
30	25	150
35	17	167
40	8	175
45	5	180

$$m = \frac{N+1}{2} = \frac{180+1}{2} = \frac{181}{2}$$

$$= 90.5$$

$$m = 25.$$

iii) mode:-

The highest frequency is 50

∴ The mode of the given distribution is 25

2) Find mean, median, mode from the following data

Profit (in lakhs)	4-7	8-11	12-15	16-19	20-23	24-27	28-31
No of companies	6	10	18	30	15	12	10
					32-35	36-39	
					6	2	

Profit (in lakhs)	No of workers	Class intervals	x	fx	cf
4-7	6	3.5-7.5	5.5	33	6
8-11	10	7.5-11.5	9.5	95	16
12-15	18	11.5-15.5	13.5	243	34
16-19	30	15.5-19.5	17.5	525	64
20-23	15	19.5-23.5	21.5	322.5	79
24-27	12	23.5-27.5	25.5	306	91
28-31	10	27.5-31.5	29.5	295	101
32-35	6	31.5-35.5	33.5	201	107
36-39	2	35.5-39.5	37.5	75	109

$$= 109$$

$$= 2095.5$$

$$CF = \frac{\text{lower limit} - \text{upper limit}}{2}$$

$$= \frac{8-7}{2} = \frac{1}{2} \quad \boxed{CF = 0.5}$$

$$\text{mean, } \bar{x} = \frac{\sum fx}{N}$$

$$\bar{x} = \frac{2095.5}{109}$$

$$\bar{x} = \underline{\underline{19.22}}$$

$$\text{median} = \frac{N}{2} = \frac{109}{2} = 54.5$$

$$\text{Now, } L = 16$$

$$Pcf = 34$$

$$f = 30$$

$$\frac{N}{2} = 54.5$$

$$i = 4$$

$$\text{median, } m = L + \left(\frac{\frac{N}{2} - Pcf}{f} \right) \times i$$

$$m = 16 + \left(\frac{54.5 - 34}{30} \right) \times 4$$

$$= 16 + \left(\frac{20.5}{30} \right) \times 4$$

$$= 16 + 0.68 \times 4$$

$$= 16 + 2.72$$

$$\boxed{m = 18.72}$$

$$\text{mode, } L = 16$$

$$f_1 = 30$$

$$f_0 = 18$$

$$f_2 = 15$$

$$Z = L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times i$$

$$= 16 + \left(\frac{30 - 18}{2(30) - 18 - 15} \right) \times 4$$

$$= 16 + \left(\frac{12}{27} \right) \times 4$$

$$Z = 16 + 0.44 \times 4$$

$$\boxed{Z = 17.76}$$

3) In a distribution the mean is 42.8 & the median is 46, find the mode using empirical relation

Sol:-

$$\begin{aligned}\text{mode} &= 3(\text{median}) - 2(\text{mean}) \\ &= 3(46) - 2(42.8) \\ &= 138 - 85.6 \\ \text{mode} &= 52.4//\end{aligned}$$

4) The mean of 150 observations was found to be 62 later it was found the 2 values were misread as 48 & 75 instead of 84 & 57 find the correct mean

Sol:-

Given, no of observations = 150

wrong mean = 62

wrong values = 48 + 75 = 123

correct mean values = 84 + 57 = 141

$$\begin{aligned}\text{Correct mean} &= \frac{(\text{wrong mean} \times \text{no of observation}) - (\text{wrong values}) + (\text{correct mean})}{\text{total no of observation}} \\ &= \frac{(62 \times 150) - 123 + 141}{150} \\ &= \frac{9300 - 123 + 141}{150} \\ &= \frac{9177 + 141}{150} \\ &= \frac{9318}{150} \\ \text{Correct mean} &= 62.12//\end{aligned}$$